

CSE110 Fall 2025 Lab Exam - 01 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class SetA {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter units consumed: ");
        int units = sc.nextInt();

        System.out.print("Enter number of days payment is late: ");
        int daysLate = sc.nextInt();

        double baseBill = 0;
        if (units <= 100) {
            baseBill = units * 5;
        } else if (units <= 300) {
            baseBill = (100 * 5) + (units - 100) * 7;
        } else {
            baseBill = (100 * 5) + (200 * 7) + (units - 300) * 10;
        }

        double penalty = 0;
        if (daysLate > 0) {
            if (daysLate <= 14) {
                penalty = baseBill * 0.10;
            } else {
                int extraDays = daysLate - 14;
                penalty = baseBill * (0.10 + (extraDays * 0.02));
            }
        }

        double totalBill = baseBill + penalty;

        System.out.printf("Total Payable: %.2f BDT%n", totalBill);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Proper import and java syntax	1
2	Creating scanner object and taking inputs and prompts properly	1
3	Checking conditions and calculate base bill properly	3
4	Checking conditions and calculate penalty properly	3
5	Calculate the Total bill and print the output properly	2

[SET-B]

SOLUTION

```
import java.util.Scanner;

public class SetB {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter current salary: ");
        double salary = sc.nextDouble();

        System.out.print("Enter years of service: ");
        int years = sc.nextInt();

        double promotion = 0;
        if (salary <= 50000) {
            promotion = salary * 0.05;
        } else if (salary <= 100000) {
            promotion = salary * 0.07;
        } else {
            promotion = salary * 0.10;
        }

        double bonus = 0;
        if (years > 0) {
            if (years <= 5) {
                bonus = salary * 0.05;
            } else {
                int extraYears = years - 5;
                bonus = salary * (0.05 + (extraYears * 0.01));
            }
        }

        double totalSalary = salary + promotion + bonus;

        System.out.printf("Total New Salary: %.2f BDT%n", totalSalary);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Proper import and java syntax	1
2	Creating scanner object and taking inputs and prompts properly	1
3	Checking conditions and calculate promotion salary properly	3
4	Checking conditions and calculate bonus salary properly	3
5	Calculate the Total salary and print the output properly	2